

Problem1

April 19, 2026

1 AA 203 — Homework 1

1.1 Problem 1.1 — Gradient descent and line search

1.1.1 a)

Using the NOC Theorem, given that f is twice differentiable, $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive semi-definite for any candidate local minimum x^* . Since $f(x) \in C^2$,

$$\nabla f(x^*) = Qx^* - b = 0 \quad \text{and} \quad \nabla^2 f(x^*) = Q \rightarrow (\text{positive-definite matrix})$$

$$\Rightarrow \boxed{x^* = Q^{-1}b}$$

Let's use Q is symmetric positive-definite $\Rightarrow$ this also implies x^* is actually a strict local minimum.

Prove x^* is a global minimum - f is defined on $\mathbb{R}^n$ that is open - $\forall x \in \mathbb{R}^n$, $\nabla^2 f(x) = Q$ is positive-definite $\Rightarrow f$ is strictly convex - Therefore the NOC is actually a necessary and sufficient condition, proving $x^* = Q^{-1}b$ is a unique global minimum.

1.1.2 b) Using Newton method

$$\begin{aligned} x^{(k+1)} &= x^{(k)} - [\nabla^2 f(x^{(k)})]^{-1} \nabla f(x^{(k)}) \\ &= x^{(k)} - Q^{-1}(Qx^{(k)} - b) \\ &= x^{(k)} - x^{(k)} + Q^{-1}b \\ &= Q^{-1}b = x^* \end{aligned}$$

$\Rightarrow$ we get to x^* in one step.

The downside is that we need to compute Q^{-1} which on average is $\mathcal{O}(n^3)$ in time and $\mathcal{O}(n^2)$ in memory. For $n \gg 10^4$, which can be very computationally challenging. Also this inversion of Q is unstable, which is only possible due to the assumed structure of the matrix (symmetric definite positive).

1.1.3 c)

$$\|Sx\|_2 = \|U\Sigma U^\top x\|_2 = \|\Sigma U^\top x\|_2 \quad (\text{because } U \text{ is orthogonal}), \quad \forall x \in \mathbb{R}^n.$$

$$\forall z \in \mathbb{R}^n. \Sigma z = \begin{pmatrix} \mu_1 & & \\ & \ddots & \\ & & \mu_n \end{pmatrix} \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} = \begin{pmatrix} \mu_1 z_1 \\ \vdots \\ \mu_n z_n \end{pmatrix}$$

Since for all i , $\mu_i^2 \leq (\max_{j \in \{1, \dots, n\}} |\mu_j|)^2$, we get

$$\|\Sigma z\|_2 = \sqrt{\sum_i \mu_i^2 z_i^2} \leq \max_{i \in \{1, \dots, n\}} |\mu_i| \sqrt{\sum_i z_i^2} = \max_{i \in \{1, \dots, n\}} |\mu_i| \|z\|_2$$

$$\Rightarrow \|\Sigma z\|_2 \leq \max_{i \in \{1, \dots, n\}} |\mu_i| \|z\|_2$$

$\forall x \in \mathbb{R}^n$,

$$\|Sx\|_2 = \|\Sigma U^\top x\|_2 \leq \max_{i \in \{1, \dots, n\}} |\mu_i| \|U^\top x\|_2 = \max_{i \in \{1, \dots, n\}} |\mu_i| \|x\|_2 \quad \text{because } U^\top \text{ is orthogonal}$$

$$\text{Thus } \boxed{\|Sx\|_2 \leq \max_{i \in \{1, \dots, n\}} |\mu_i| \|x\|_2}$$

1.1.4 d)

Q is a symmetric, definite positive matrix $\rightarrow$ spectral theorem implies

$$\exists U \in \mathbb{R}^{n \times n} \text{ orthogonal s.t. } Q = U \Sigma_Q U^\top \quad \text{where } \Sigma_Q = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Note $e_i = (0, \dots, \underset{\substack{\uparrow \\ \text{ith pos}}}{1}, \dots, 0)^\top$ the canonical base of $\mathbb{R}^n$, and $v_i = U e_i$.

$$\begin{aligned} (I - \eta Q) v_i &= (I - \eta Q) U e_i \\ &= U e_i - \eta U \Sigma_Q U^\top U e_i \\ &= U e_i - \eta U \Sigma_Q e_i \\ &= U e_i - \eta \lambda_i U e_i \\ &= (1 - \eta \lambda_i) U e_i \\ &= (1 - \eta \lambda_i) v_i \quad (**) \end{aligned}$$

Thus $(I - \eta Q) v_i = (1 - \eta \lambda_i) v_i$, and (v_i) , $i = 1, \dots, n$ is another base of $\mathbb{R}^n$ since U is orthogonal.
(**)

For each i , $(1 - \eta \lambda_i)$ is an eigenvalue of $I - \eta Q$ due to (**).

1.1.5 e)

For $k \geq 0$,

$$\begin{aligned}
\delta^{(k+1)} &= \|x^{(k+1)} - x^*\|_2 \\
&= \|x^{(k)} - \eta(Qx^{(k)} - b) - \underbrace{[x^* - \eta(Qx^* - b)]}_{=0}\|_2 \quad (\text{since } \nabla f(x^*) = Qx^* - b = 0) \\
&= \|[I - \eta Q]x^{(k)} + \eta b - [I - \eta Q]x^* - \eta b\|_2 \\
&= \|[I - \eta Q][x^{(k)} - x^*]\|_2 \\
&\leq \max_{i \in \{1, \dots, n\}} |1 - \eta \lambda_i| \cdot \|x^{(k)} - x^*\|_2
\end{aligned}$$

using question c) (Note that since Q is symmetric “ $S = I - \eta Q$ ” is also symmetric, so c) applies in here, and in d) we computed the eigen values).

$$\implies \delta^{(k+1)} \leq \gamma(\eta) \delta^{(k)}$$

By recursion, since all these are positive we have:

$$\forall k \geq 0, \quad \delta^{(k)} \leq \gamma(\eta)^k \cdot \underbrace{\delta^{(0)}}_{= \|x^{(0)} - x^*\|_2 = \delta_0}$$

1.1.6 f)

$d^{(k)} = -\nabla f(x^{(k)})$. Define $g(\eta) = f(x^{(k)} + \eta d^{(k)})$.

$$\begin{aligned}
g'(\eta) &= (x^{(k)})^\top Q d^{(k)} + \eta d^{(k)\top} Q d^{(k)} - b^\top d^{(k)} \\
&= d^{(k)\top} (Qx^{(k)} - b) + \eta d^{(k)\top} Q d^{(k)}
\end{aligned}$$

Let's find η^* : η unconstrained optimization problem, so

$$\min_{\eta \geq 0} g(\eta)$$

since we have a 1D convex problem defined on the positive reals, a necessary and sufficient condition is

$$g'(\eta) = 0$$

$$0 = d^{(k)\top} (Q(x^{(k)} + \eta d^{(k)}) - b) = d^{(k)\top} [(Qx^{(k)} - b) + \eta Qd^{(k)}]$$

$$0 = d^{(k)\top} (-d^{(k)}) + \eta d^{(k)\top} Q d^{(k)} \quad \left(= -\underbrace{\|d^{(k)}\|_2^2} \right)$$

$$\eta^* = \frac{\|d^{(k)}\|_2^2}{d^{(k)\top} Q d^{(k)}}$$

η^* happens to be the global minimum of g defined on $\mathbb{R}$, hence it is also a global minimum of g on $\mathbb{R}^+$.

$$\Rightarrow \eta^{(k)} = \arg \min_{\eta \geq 0} f(x^{(k)} + \eta d^{(k)}) = \frac{\|d^{(k)}\|_2^2}{d^{(k)\top} Q d^{(k)}}$$

1.1.7 g)

$$f(x) = \frac{1}{2} x_1^2 + \gamma x_2^2 \cdot \frac{1}{2} = \frac{1}{2} x^\top Q x - b^\top x$$

For

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & \gamma \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Please see code.

```
[1]: import numpy as np
import matplotlib.pyplot as plt

def gd_const(Q, x0, eta, n_iters=60):
    xs = [np.asarray(x0, dtype=float)]
    for _ in range(n_iters):
        xs.append(xs[-1] - eta * (Q @ xs[-1]))
    return np.array(xs)

def gd_linesearch(Q, x0, n_iters=60, tol=1e-12):
    xs = [np.asarray(x0, dtype=float)]
    for _ in range(n_iters):
        d = -Q @ xs[-1]
        if np.linalg.norm(d) < tol:
            break
        eta = (d @ d) / (d @ Q @ d)
        xs.append(xs[-1] + eta * d)
    return np.array(xs)

def plot_traj(ax, Q, traj, title, lim=6):
    g = np.linspace(-lim, lim, 300)
    X, Y = np.meshgrid(g, g)
    Z = 0.5 * (Q[0, 0] * X**2 + Q[1, 1] * Y**2)
    ax.contour(X, Y, Z, levels=25, cmap="viridis", alpha=0.6)
    ax.plot(traj[:, 0], traj[:, 1], "r.-", lw=1, ms=4)
    ax.plot(0, 0, "k*", ms=12)
    ax.set_title(title)
    ax.set_xlabel(r"$x_1$"); ax.set_ylabel(r"$x_2$")
    ax.set_aspect("equal")
    ax.set_xlim(-lim, lim); ax.set_ylim(-lim, lim)

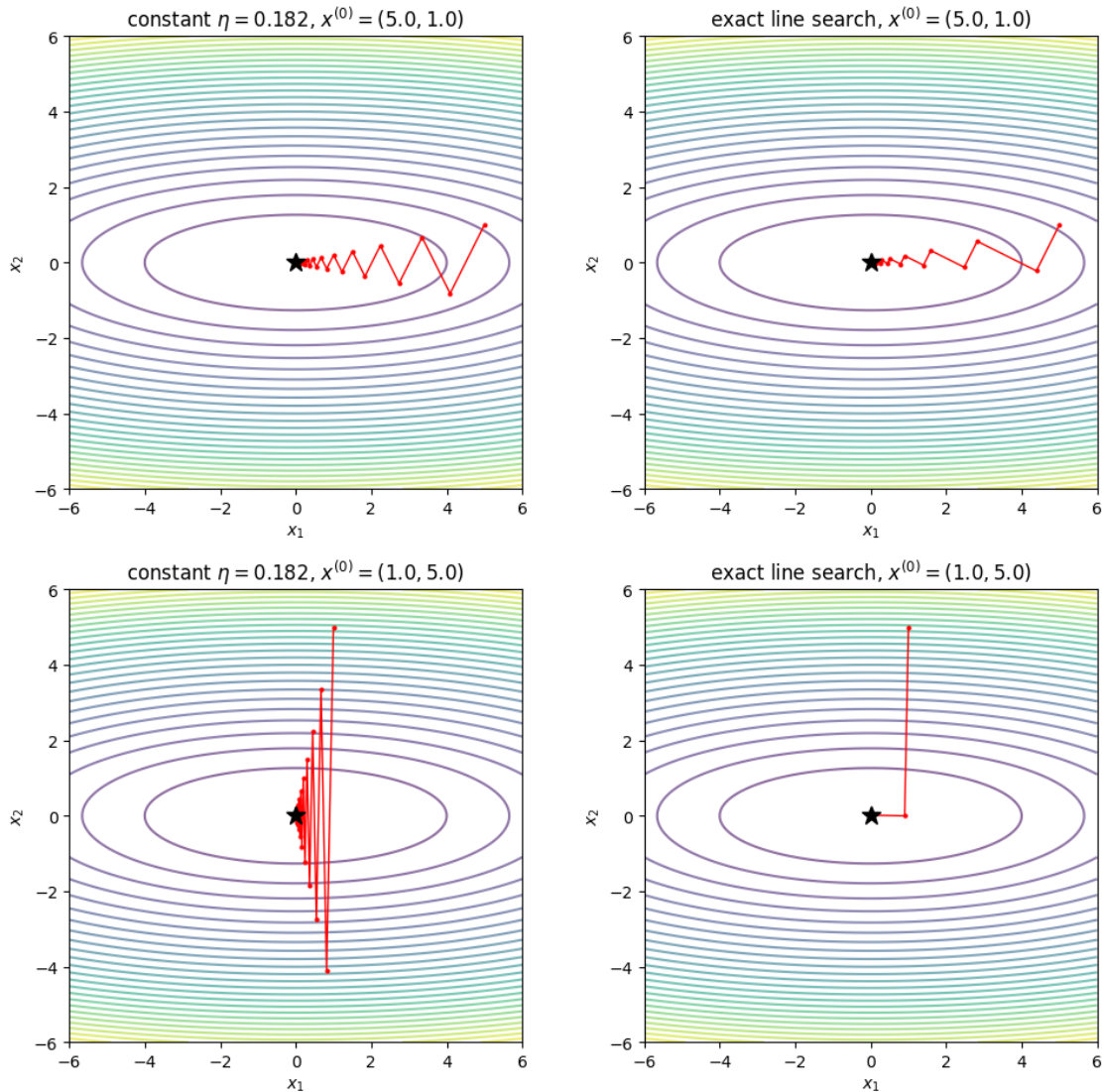
starts = [(5.0, 1.0), (1.0, 5.0)]
```

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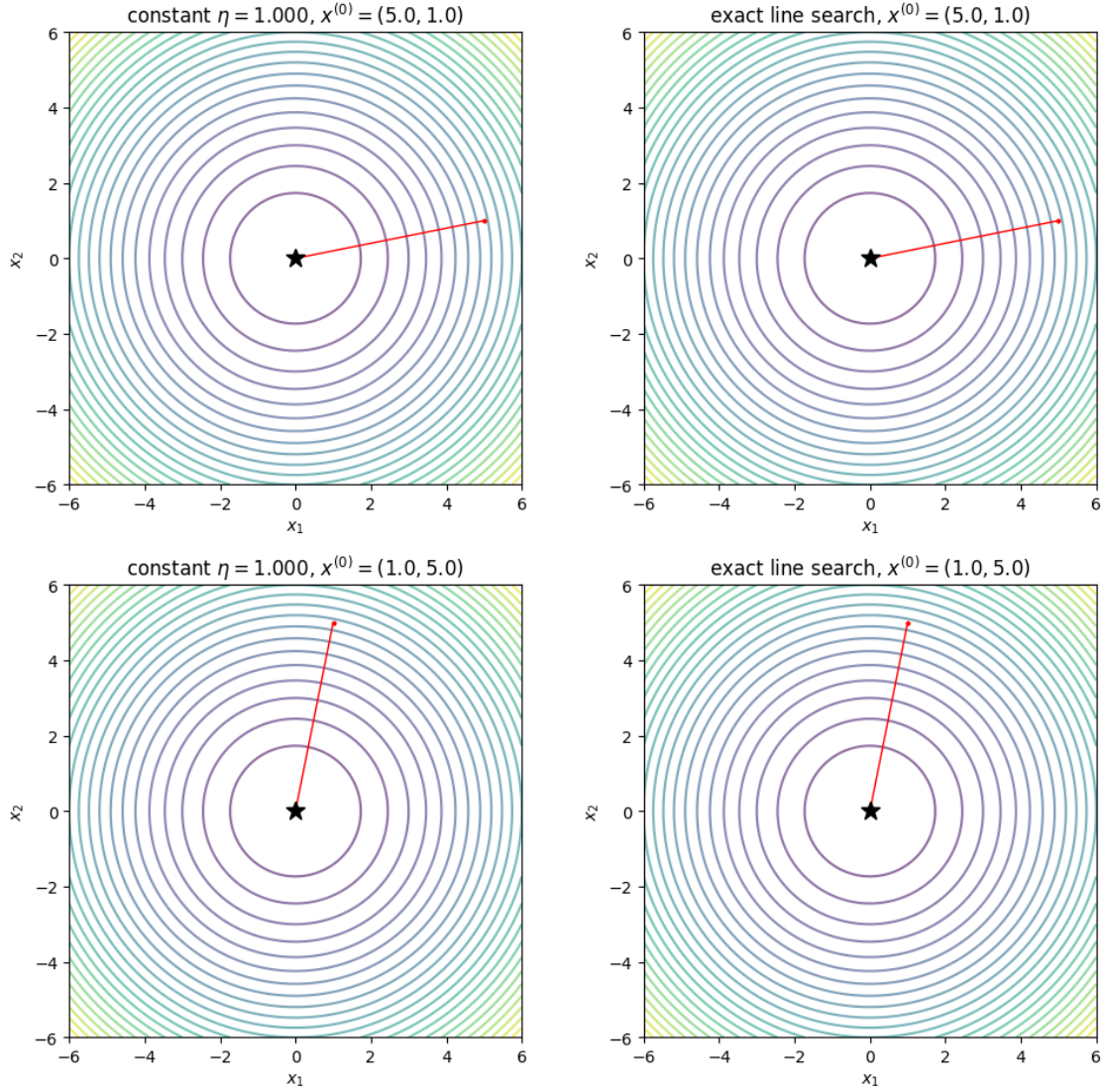
for gamma in (10, 1):
    Q = np.diag([1.0, float(gamma)])
    eta = 2.0 / (1.0 + gamma) # 2 / (lambda_min + lambda_max)
    fig, axes = plt.subplots(2, 2, figsize=(10, 10))
    for i, x0 in enumerate(starts):
        tc = gd_const(Q, x0, eta=eta)
        tl = gd_linesearch(Q, x0)
        plot_traj(axes[i, 0], Q, tc, f"constant  $\eta={eta:.3f}$ ,  $x^{(0)}={x0}$ ")
        plot_traj(axes[i, 1], Q, tl, f"exact line search,  $x^{(0)}={x0}$ ")
    fig.suptitle(fr"$\gamma = {gamma}$", fontsize=14)
    plt.tight_layout()
    plt.show()

```

$\gamma = 10$



$\gamma = 1$



Observations.

Exact line search always reaches $x^* = (0, 0)$. For $\gamma = 10$ the iterates zig-zag: consecutive line-search directions are orthogonal (since $\nabla f(x^{(k+1)})^\top d^{(k)} = g'(\eta^{(k)}) = 0$), and the bigger the condition number γ , the more severe the zig-zag. For $\gamma = 1$ the level sets are circles, the gradient points straight at x^* , and we converge in a single iteration from any starting point.

Constant step size needs $\eta < 2/\lambda_{\max} = 2/\gamma$ to be stable (from (e), $\gamma(\eta) < 1$). The best constant step is $\eta^* = 2/(\lambda_{\min} + \lambda_{\max}) = 2/(1 + \gamma)$, with contraction factor $(\gamma - 1)/(\gamma + 1)$. For $\gamma = 10$ that is $9/11 \approx 0.82$, so progress is slow and we see the same zig-zag as with exact line search. If η is too large the iteration diverges; too small and progress crawls. For $\gamma = 1$, $\eta^* = 1$ also gives one-step

convergence.